

Donavan Borges

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EDUCATION

• California Institute of Technology (Caltech)

B.S. in Computer Science (Robotics Track)

Expected December, 2026

Pasadena, CA

- **Relevant Coursework:** Experimental Robotics; Robotics; Machine Learning; Algorithms

EXPERIENCE & PROJECTS

• Autonomous Combat Robot Lead

Sep 2025 – Present

Caltech Robotics Team

- Leading Caltech's first autonomous combat robotics project, integrating YOLO models trained in Roboflow with an external Intel RealSense camera for perception.
- Developing the autonomy stack for real-time detection, tracking, embedded control, and decision-making; validating algorithms in simulation.
- Building the communication pipeline between a laptop-based controller and ESP32-based robot hardware with controls and hardware subteams.

• Autonomous Multi-Robot Systems

Mar 2025 – Jun 2025

ME/CS/EE 129: Experimental Robotics

- Programmed mobile robots for exploration, mapping, navigation, and reactive obstacle avoidance using SLAM and Dijkstra's Algorithm.
- Designed multi-threaded ROS architectures on Raspberry Pi 4 and implemented real-time motor control with IR and ultrasound sensing.
- Built planning and decision modules for a final multi-robot autonomous demonstration.

• Monte Carlo Simulation

Jan 2026 - Mar 2026

ME/CS/EE 133b

- Implemented a particle-filter-based Monte Carlo Localization pipeline using probabilistic motion updates, simulated 2D LiDAR ray casting, Gaussian weighting, and adaptive resampling.
- Evaluated localization across four map types by varying particle count, LiDAR ray count, measurement noise, and motion noise.
- Analyzed tradeoffs between accuracy, convergence, and effective sample size, highlighting how ambiguity and parameter tuning affect robustness.

• StackBot: Autonomous Cup Stacking Robot

Jan 2026 - Mar 2026

ME/CS/EE 134

- Built a 5-DOF robotic arm that autonomously constructs a cup pyramid from human-placed ArUco markers, then downstacks it into a single stack.
- Integrated overhead Intel RealSense sensing, ArUco-based calibration, HSV color filtering, and multi-depth-layer cup detection for cup localization.
- Developed a ROS architecture with mapping, planning, and motor-control nodes plus pyramid-state modeling and gripper feedback for replanning and grasp recovery.

SKILLS

- **Programming & Simulation:** Python, C++, MATLAB, ROS2; Linux; RViz, Isaac Sim, MuJoCo, Unreal Engine, Unity; multithreaded programming
- **Robotics & Systems:** Motor control; sensor integration; feedback control; planning & navigation
- **Hardware Platforms:** Embedded systems, Arduino, Raspberry Pi, ESP32; autonomous mobile robots
- **Other:** Team leadership; project planning; fast iteration; adaptability; communication

LEADERSHIP EXPERIENCE

• Treasurer & Business/Software Team Lead

Apr 2025 – Present

Caltech Robotics Team

- Managed finances for multi-project robotics initiatives; raised over \$15K in one term toward an annual \$50K goal.
- Led software development across quadruped, combat robotics, RoboSub, and website efforts and trained new members on ROS, Git, and robotics workflows.
- Built sponsor relationships, drafted proposals, and coordinated competitions, lab tours, and STEM outreach.

AWARDS

• QuestBridge National College Match Scholarship

Dec 2020

QuestBridge

- Full four-year scholarship awarded for outstanding academic achievement and leadership potential among high-achieving, low-income students nationwide